

Intermediate Macro

Chapter 23: Monetary Policy (A Summing up)

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1 Chapter 23: Monetary Policy (A Summing up)

23.1 What We Have Learned

23.2 From Money Targeting to Inflation Targeting

23.3 The Optimal Inflation Rate

23.4 Unconventional Monetary Policy

23.5 Monetary Policy and Financial Stability



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- Chapter 23 introduces **monetary aggregates**. **M1** is the sum of currency and checkable deposits. **M2** is the sum of **M1**, money market mutual fund shares, money market deposit accounts, and time deposits.
- The **Taylor rule** argues that central banks should deviate from their target interest rates to the extent that they care about short-run fluctuations in inflation and unemployment.
- **Quantitative easing** is presented as an unconventional monetary policy tool.
- **Macro prudential tools** are rules aimed directly at borrowers or lenders, or banks and other financial institutions as the case may require.

23.1 What We Have Learned

- In the short run, expansionary monetary policy lowers interest rates and causes the exchange rate to depreciate.
- As a result, output and the price level increase.
- In the medium and long run, money growth determines inflation, but has no effect on output.
- Thus, the medium-run inflation rate is determined by the medium-run rate of growth of the money supply.
- With this in mind, the Federal Reserve has to choose a rate of inflation to target in the medium run and to decide when and by how much to deviate from this target in the short run.

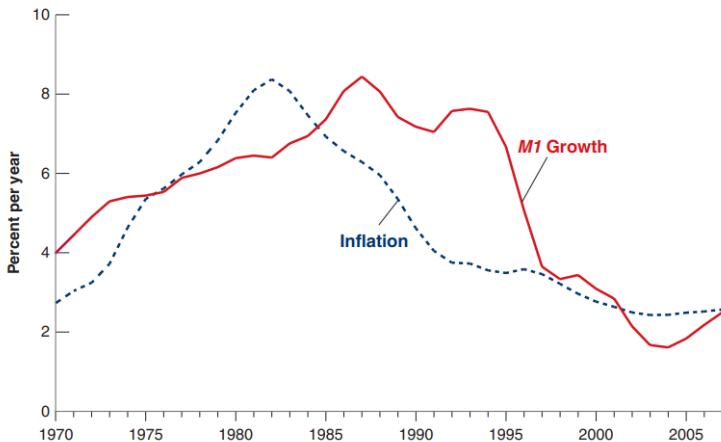
Money Targeting

23.2 From Money Targeting to Inflation Targeting

- One can think of the goals of monetary policy as twofold:
 - ① First, to maintain low and stable inflation
 - ② Second, to stabilize output around potential
- Until the 1990s, central bankers typically thought about monetary policy in terms of the rate of money growth.
- A central bank chose a rate of money growth that corresponds to its inflation target, and worried about how closely the growth of monetary aggregates matched its goals.
- But it turns out that money demand does not have a stable relationship with any of the monetary aggregates.
- As a result, inflation does not have a tight relationship with the growth of any particular monetary aggregate.

M1 Growth and Inflation: 10-Year Averages, 1970 to 2007

23.2 From Money Targeting to Inflation Targeting



Inflation Targeting

23.2 From Money Targeting to Inflation Targeting

- Faced with this reality, most central banks now think about monetary policy in terms of the nominal interest rate.
- A central bank tries to achieve its target inflation rate by changing the interest rate.
- Focusing exclusively on inflation would seem to eliminate any role for monetary policy in reducing output fluctuations. But, in fact, this is not the case.
- To see why, recall the Phillips curve relation (equation (8.10)):

$$\pi_t = \pi_t^e - \alpha(u_t - u_n)$$

- Let the inflation target be $\bar{\pi}$ and assume that this target is credible, so that people expect inflation to be equal to the target. The relation becomes:

$$\pi_t = \bar{\pi} - \alpha(u_t - u_n)$$

Divine Coincidence

23.2 From Money Targeting to Inflation Targeting

- Note that, if the central bank can hit its inflation target exactly, so $\pi = \bar{\pi}$, unemployment will be equal to its natural rate.
- Even if policymakers did not care about inflation per se but cared only about output, inflation targeting would still make sense.
- Keeping inflation stable is a way to keep unemployment at the natural rate, or equivalently to keep output at potential.
- This result has been dubbed the **divine coincidence**.

Flexible Inflation Targeting

23.2 From Money Targeting to Inflation Targeting

- The result in the previous slide is a useful benchmark, but it is too strong. Life is not that nice.
- The main objection is that, the Phillips curve relation is far from an exact relation.
- Rather than try to achieve the target inflation rate each period, at the risk of large fluctuations in unemployment, it is better to try to achieve it only over time.
- So most central banks adopted what is called **flexible inflation targeting**: When inflation is away from target, rather than trying to return it to target right away, they adjust the interest rate to return to the target inflation rate over time.

The Interest Rate Rule

23.2 From Money Targeting to Inflation Targeting

- Note that inflation is not under the direct control of the central bank. The policy rate is.
- Thus, the question is how to set the policy rate to achieve the target rate of inflation.
- The answer is a simple one: When inflation is higher than the target, increase the policy rate to decrease demand and the pressure on prices; when it is below the target rate of inflation, decrease the policy rate.
- With this in mind, in the 1990s, John Taylor suggested a rule known as the **Taylor rule**:

$$i_t = \bar{i} + a(\pi_t - \bar{\pi}) - b(u_t - u_n)$$

where a and b are positive coefficients chosen by the central bank.

Taylor Rule

23.2 From Money Targeting to Inflation Targeting

- The coefficients a and b reflect how much the central bank cares about inflation versus unemployment.
- Note that the coefficient a should be greater than one, because the real interest rate affects spending.
- Thus, when inflation increase, the central bank must increase the nominal rate by a greater amount to increase the real interest rate and reduce spending.
- Basically, the Taylor rule says that the central bank should respond to short-run fluctuations to the extent it cares about them.
- Evidence suggests that the Taylor rule describes quite well the actual behavior of the Fed and the Bundesbank over the 20 years. Then the crisis came and raised many questions.

The Costs of Inflation

23.3 The Optimal Inflation Rate

- The optimal medium-run inflation rate depends on the costs and benefits of inflation.
 - The costs of inflation vary with the level of inflation. Very high rates of inflation disrupt economic activity; low rates of inflation are much less costly.
 - Inflation creates four costs, all of which increase with the rate of inflation.
- ① **Shoe-Leather Costs:** higher inflation leads to higher nominal interest rates, which cause people to economize on holding of money. This effort requires resources, referred to as shoe-leather costs. Such costs can become quite large during hyperinflation, but are unlikely to be very important when inflation is low.

Tax Distortions and Money Illusion

23.3 The Optimal Inflation Rate

- ② **Tax Distortions:** when the tax system is not indexed, the real tax burden will depend on the rate of inflation. In the United States, this issue applies in particular to nominal interest income and capital gains. The tax distortions created by inflation affect the allocation of economic resources. Indexing the tax code to account for inflation could address these issues.
- ③ **Money Illusion:** with inflation, real and nominal values diverge. Evidence suggests that people suffer from money illusion, i.e., they make systematic errors in assessing nominal versus real changes. Thus, inflation leads people and firms to make incorrect decisions.

Inflation Variability

23.3 The Optimal Inflation Rate

- ④ **Inflation Variability:** because higher inflation is associated with more variable inflation, nominal assets become riskier in real terms. This cost could be avoided through increased reliance on indexed bonds (bonds whose nominal payments are indexed to the inflation rate). A number of countries, including the United States, now issue indexed bonds.

The Benefits of Inflation

23.3 The Optimal Inflation Rate

- Inflation also has three benefits.
- ① **Seignorage:** Seignorage—the amount of resources the government collects by printing money—increases at higher rates of inflation, at least over some initial range. But the amount of seignorage available at low rates of inflation in countries with otherwise highly developed tax systems is too small to make this a powerful argument for low—as opposed to zero—rates of inflation.
- ② **Money Illusion Revisited:** money illusion may cause workers to resist nominal wage cuts, independent of the effects on the real wage. As a result, ongoing inflation may be necessary to achieve the real wage cuts that are sometimes required to reallocate labor across sectors.

The Option of Negative Real Interest Rates

23.3 The Optimal Inflation Rate

- ③ **The Option of Negative Real Interest Rates:** since the nominal interest rate cannot fall below zero, inflation makes it possible for governments to achieve negative real interest rates, an option that may be useful to stimulate output during a recession.

The Optimal Inflation Rate: The State of the Debate

23.3 The Optimal Inflation Rate

- At present, the debate over the optimal inflation rate is between those who favor low inflation (say 3% or so) and those who favor zero inflation.
- Those who favor low inflation argue that the costs of inflation are low in this range and that some of these costs could be avoided by indexing the tax system and issuing more indexed bonds.
- In addition, there are benefits to maintaining some inflation, and the cost of eliminating it (in terms of higher unemployment for a time) may outweigh any future benefits.
- Those who favor zero inflation argue that such a target corresponds to price stability, which simplifies decision-making and helps alleviate the time inconsistency problem by establishing a simple and clear goal for monetary policy.

Quantitative Easing

23.4 Unconventional Monetary Policy

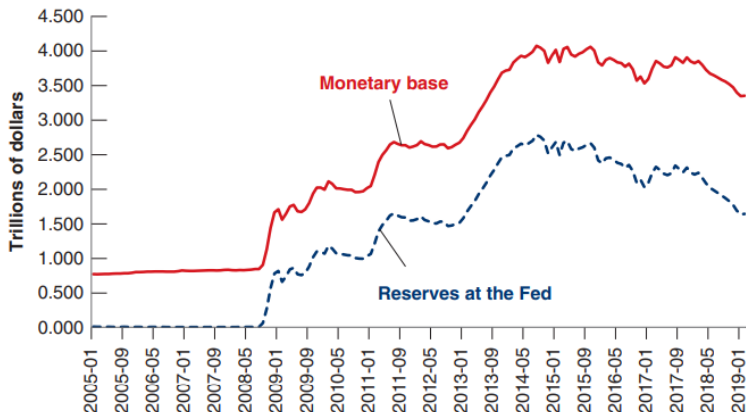
- When, at the start of the crisis, the interest rate reached the zero lower bound, central banks were unable to decrease it further and thus lost the use of **conventional monetary policy**.
- Central banks explored other ways to affect activity; a set of measures known as **unconventional monetary policy**.
- For example, central banks began buying financial assets other than short-term bonds.
- This increase in demand pushed prices higher and reduced longer-term interest rates and stimulated consumption and investment.
- These asset purchase programs became known as **quantitative easing**(QE), or **credit easing**.
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23.4 Unconventional Monetary Policy

- The primary assets purchased in three separate QE rounds were mortgage backed securities and long-term U.S. Treasuries.
- The end result was lower interest rates and a massive expansion of the Fed's balance sheet.
- Currently, most of this money expansion is being held at excess bank reserves.
- The Fed's decision point is whether to return its balance sheet to pre-crisis levels or to gradually reduce asset holdings.

The Evolution of the US Monetary Base from 2005 to 2008

23.4 Unconventional Monetary Policy



Monetary Policy since the End of the Liquidity Trap

23.4 Unconventional Monetary Policy

- When the Fed decided at the end of 2015 that the economy was stronger and it should increase the federal funds rate again, it faced an obvious problem.
- If it continued to pay zero interest rate on reserves (or, more precisely, to pay a very low one), banks would not be willing to hold reserves.
- They would try to sell all of them in the federal funds market, and, unless the Fed was willing to buy back them all, the federal funds rate would go back to zero.
- If the Fed did not want to this to happen, it had to start paying interest on reserves and this is what it did, and still does today.

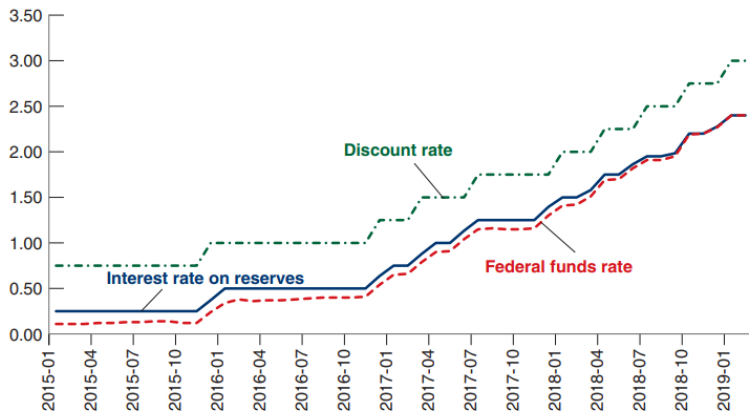
Corridor System

23.4 Unconventional Monetary Policy

- The Fed now operates a **corridor system**. It sets two rates:
 - ① the rate on reserves, the rate at which banks can in effect lend to the Fed; and
 - ② the discount rate, the rate at which banks can borrow from the Fed.
- The federal funds rate is still determined in the federal funds market, but it must be in the corridor created by the two rates.

The Interest Rate on Reserves, the Federal Funds Rate, and the Discount Rate since 2015

23.4 Unconventional Monetary Policy



23.5 Monetary Policy and Financial Stability

- Until 2007, most countries saw stable inflation, and output fluctuations that were smaller than they had been in the past.
- Many central banks believed that inflation targeting provided them with a solid framework for monetary policy.
- It is believed that the crisis has thus presented countries with the challenge of the liquidity trap and new processes to achieve macroeconomic stability beyond stable inflation.

23.5 Monetary Policy and Financial Stability

- One of the functions of the Fed is to prevent bank runs and ensure continued confidence in the banking system.
- Depositors need to feel that their money is safe and accessible.
- Deposit insurance and the Fed's readiness to loan banks funds to keep banks from liquidating assets to cover withdrawals (i.e. **lender of last resort** function).

23.5 Monetary Policy and Financial Stability

- The crisis has created a broad consensus that it is risky to wait, and it is better to prevent high leverage.
- Also, it is now believed that to deal with bubbles, credit booms, or dangerous behavior in the financial system, the interest rate is not the right policy instrument.
- Many believe the interest rate is too blunt an instrument, affecting the *whole* economy rather than resolving the problem at hand.
- An analysis of new instruments, which are called **macroprudential tools** are introduced.

Examples and Remaining Challenges

23.5 Monetary Policy and Financial Stability

- These tools are rules that are aimed *directly* at borrowers, lenders, banks and/or other financial institutions.
 - ① **loan-to-value (LTV) ration**
 - ② **Basel II and Basel III**
 - ③ **capital controls**
- There are many questions regarding how well these macroprudential tools work and their interaction with traditional monetary policy tools.
- Lastly, there is debate over whether the control should be under the central bank or a separate authority.
- The crisis has shown that macroeconomic stability requires the use not only of traditional monetary instruments but also of macroprudential tools.
- How best to use them is one of the challenges facing macroeconomic policymakers today.